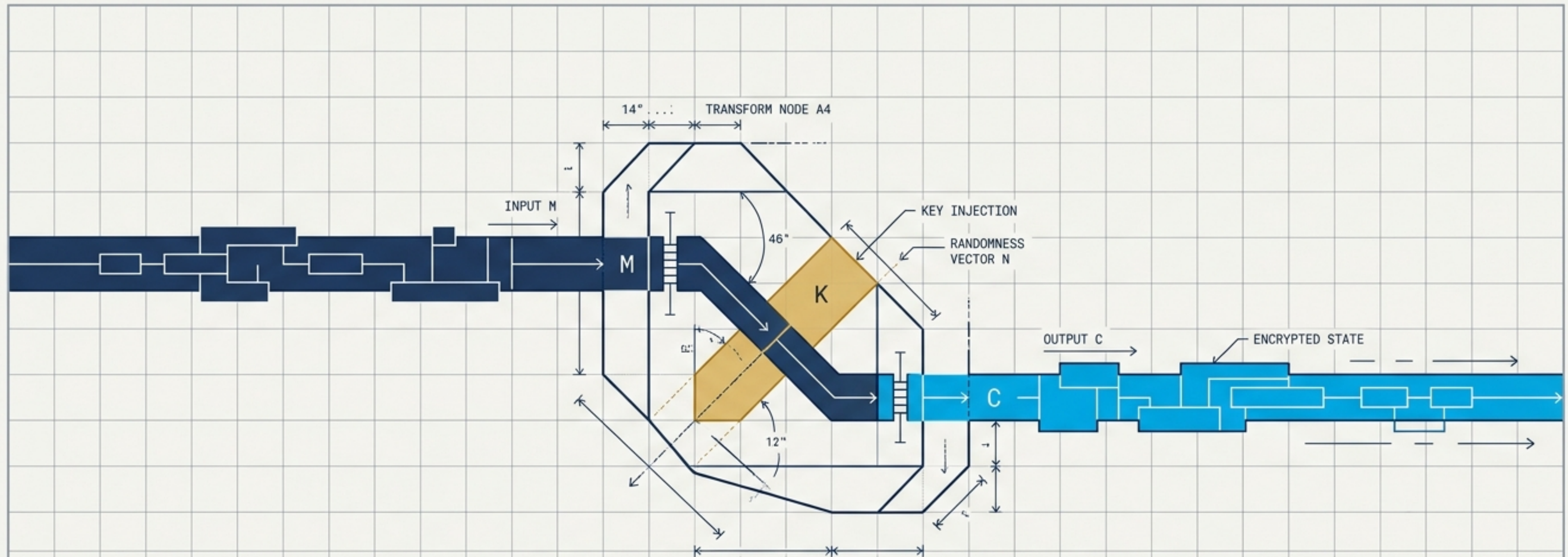




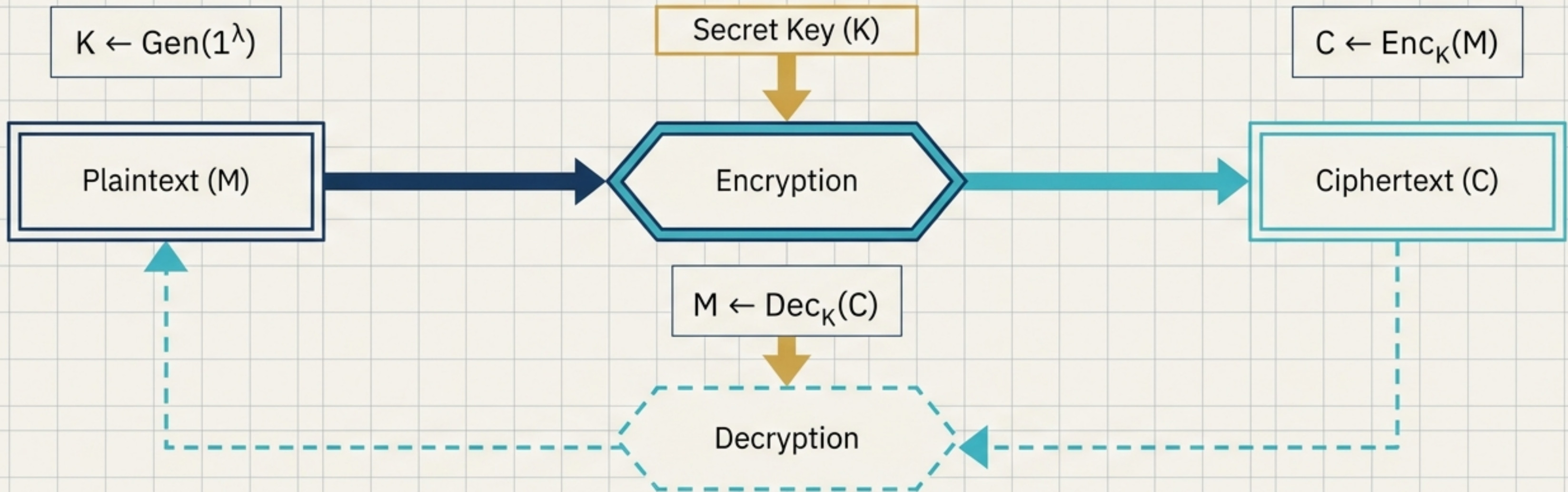
The Architecture of Secrecy

Engineering Symmetric Cryptography for the Modern Web

A systemic approach to confidentiality, from perfect secrecy to authenticated encryption.

The Principle of Symmetric Encryption

The Cryptographic Pipeline



Correctness vs. Security

The equation $\text{Dec}_K(\text{Enc}_K(M)) = M$ only proves the system functions. Security requires defining exactly what an adversary can observe and influence.

The Bottleneck: Key Distribution

Symmetric encryption secures the data, but an attacker observing the channel prevents safe key transmission. This necessitates public-key cryptography to secure the shared secret.

The Theoretical Ideal: The One-Time Pad

Perfect Secrecy

$$C = M \text{ XOR } K$$

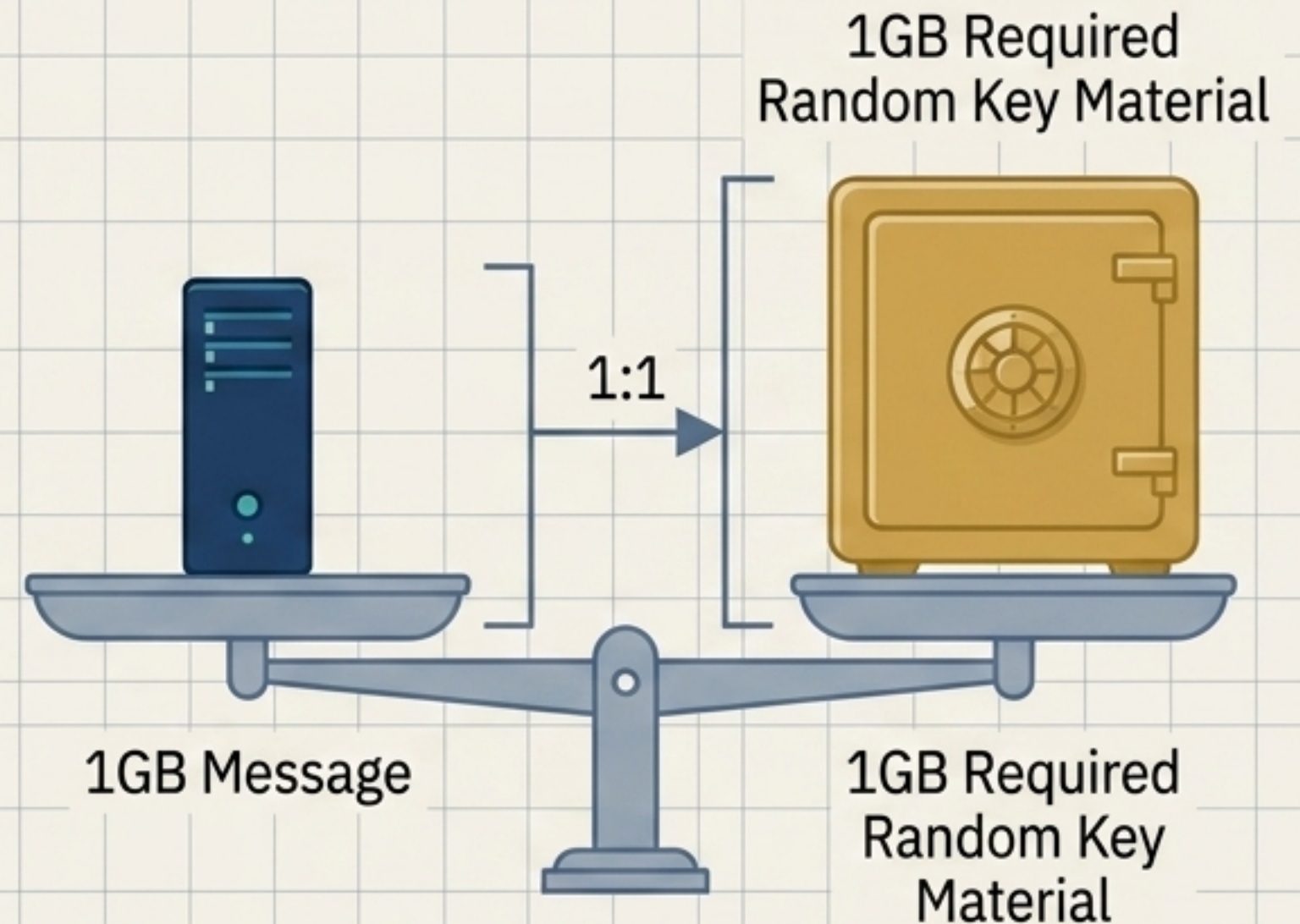
$$M = C \text{ XOR } K$$

$$\Pr[M=m \mid C=c] = \Pr[M=m]$$

- 1. Genuinely random.
- 2. Kept perfectly secret.
- 3. Length $\geq$ Message length.
- 4. NEVER REUSED.

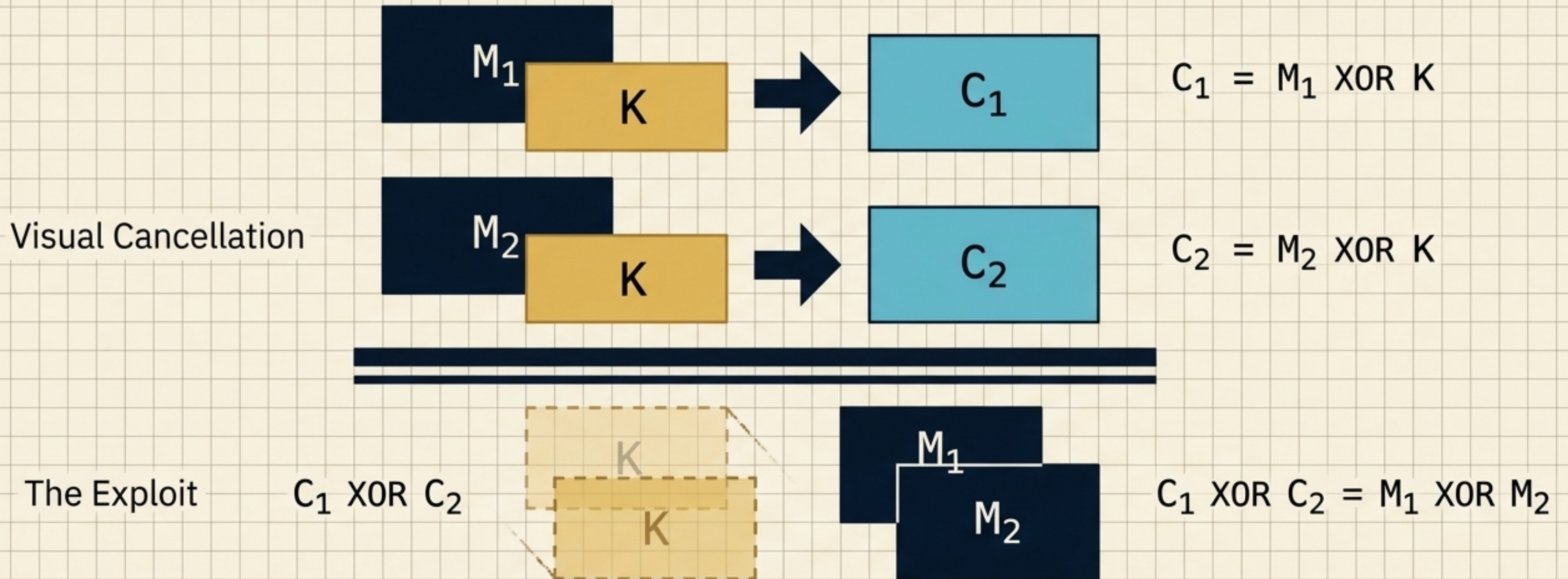
An adversary with infinite computational power learns exactly zero information about the plaintext.

The Impractical Reality



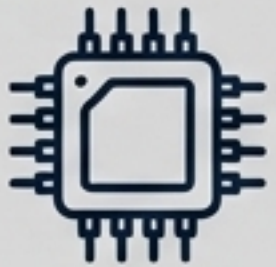
Protecting a 1GB message requires 1GB of pre-distributed, perfectly random, one-time-use key material.

The Vulnerability of State: The Two-Time Pad

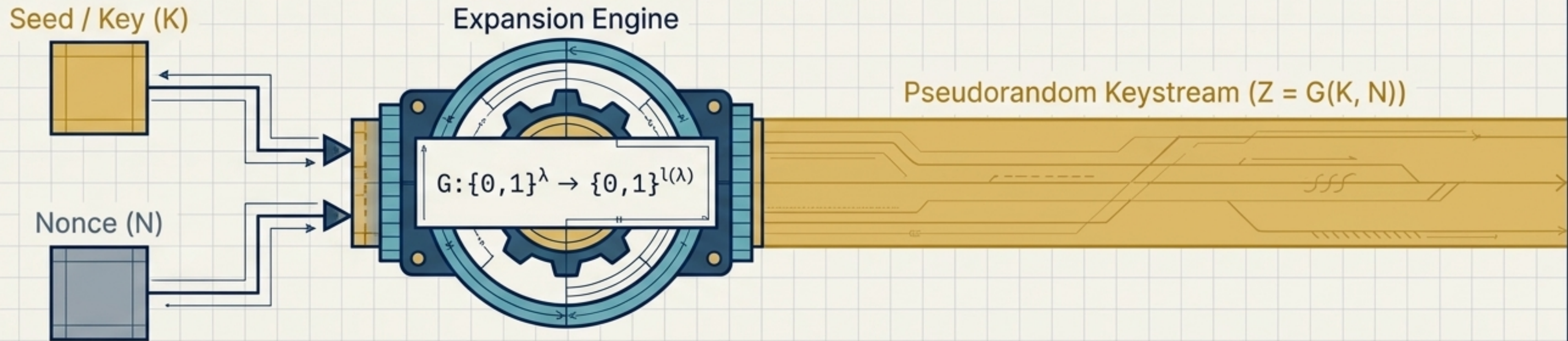


The Takeaway: The key cancels completely. Reusing a state variable—whether a One-Time Pad key, a modern nonce, or a stream cipher counter—destroys the cryptographic guarantee instantly.

The Paradigm Shift in Security

	Information-Theoretic Security (OTP)	Computational Security (Modern Cryptography)
Key Requirement	Key size must be $\geq$ Message size. 	Short, fixed-length keys (e.g., 128 or 256 bits). 
Adversary Power	Impervious to infinite computing power.	Secure against efficient, polynomial-time adversaries.
Mathematical Basis	Probability theory.	Computational complexity & indistinguishability.
Practicality	Unscalable for modern data networks.	Powers the entire modern internet infrastructure.

Solving Expansion: PRGs & The Nonce



The Mechanism

A Pseudorandom Generator (PRG) expands a short random seed into a long sequence computationally indistinguishable from true randomness.

The Role of the Nonce

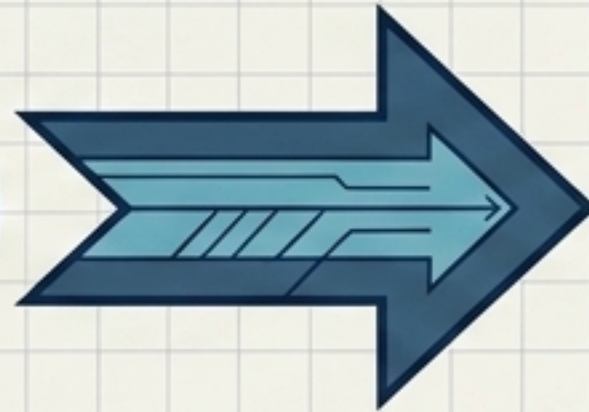
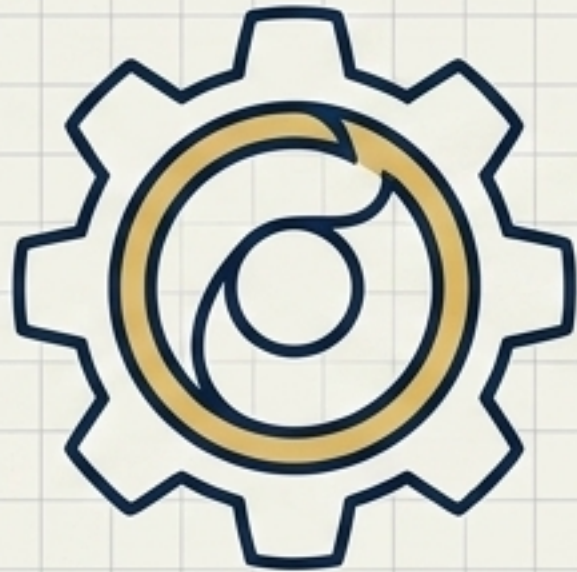
Modern encryption permits one secret key to protect many messages. By adding a Nonce (Number Used Once), $G(K, N)$ ensures different encryptions under the same key produce totally different keystreams.

Constraint

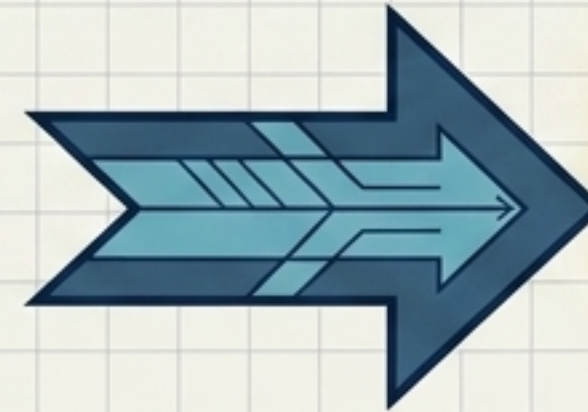
The nonce does not need to be secret, but for most stream ciphers, uniqueness is an absolute requirement.

The Evolution of Stream Ciphers

LFSR



RC4



Salsa20 / ChaCha



Status: Broken

Linear Feedback Shift Register.

Generates bits via recurrence over $GF(2)$. Highly vulnerable: linear structure allows attackers to reconstruct the recurrence (e.g., Berlekamp-Massey algorithm).



Status: Obsolete

Historically dominant due to speed and simplicity. Destroyed by cryptanalysis revealing significant statistical biases in its keystream.

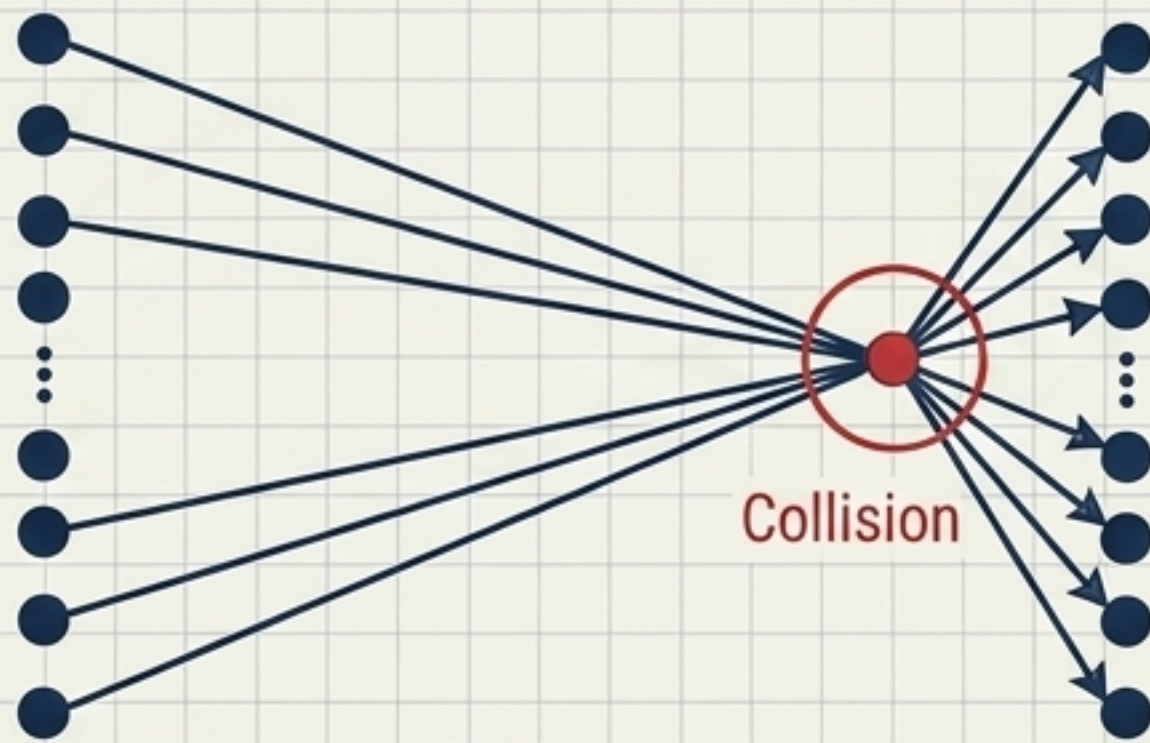


Status: Modern Standard

Uses a carefully designed combination of addition modulo 2^{32} , XOR, and rotations. Proves that simple operations can create unbreakable non-linear structures.

The Engine Blocks: PRFs and PRPs

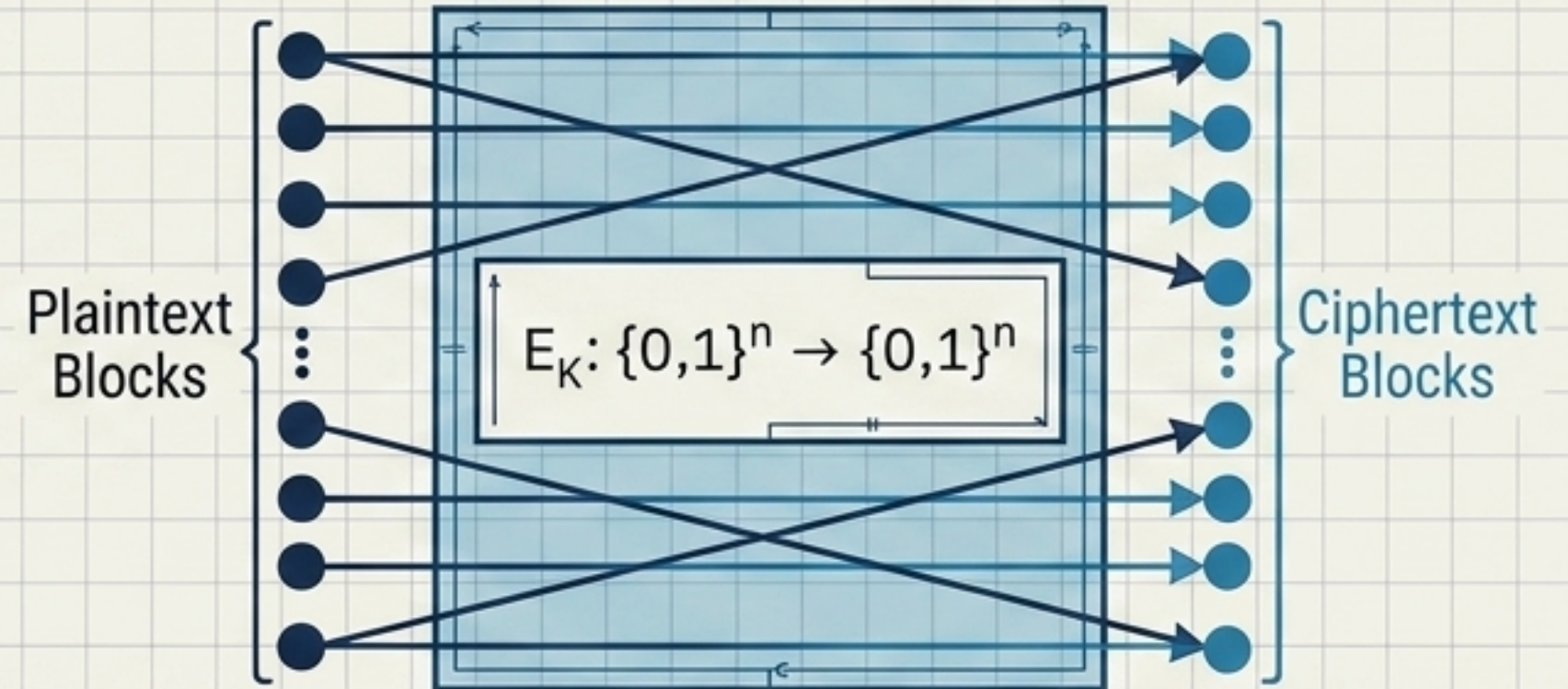
Pseudorandom Function (PRF)



A keyed function computationally indistinguishable from a true random function.

Multiple inputs can map to the same output. Collisions are theoretically possible.

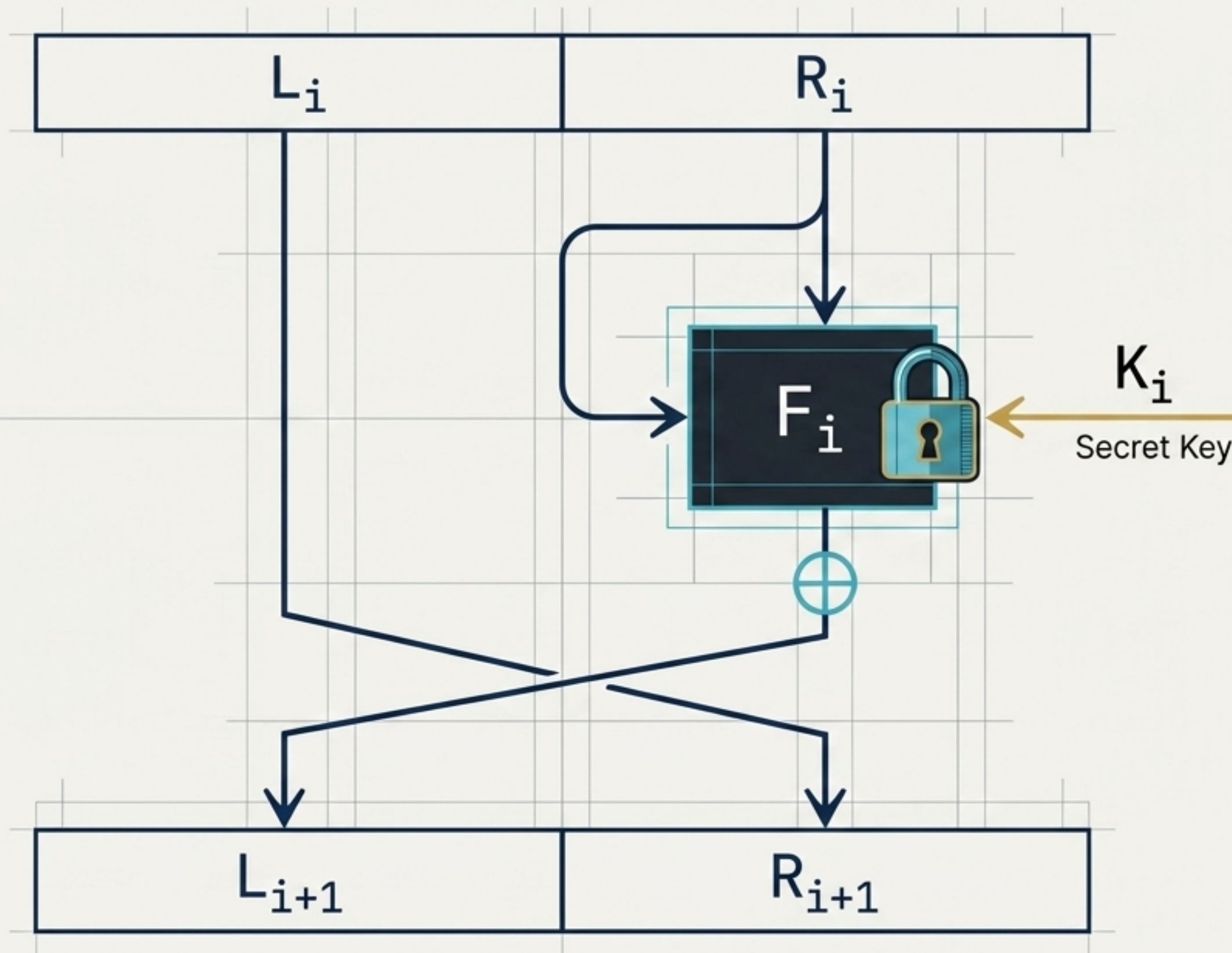
Pseudorandom Permutation (PRP) / Block Cipher



A keyed permutation indistinguishable from a random permutation. Maps fixed-size plaintext blocks to fixed-size ciphertext blocks.

$x \neq y \Rightarrow P_K(x) \neq P_K(y)$. Collisions are impossible.
Decryption is simply its inverse: $D_K = E_K^{-1}$.

The Feistel Network: Invertibility by Design



The Mechanism:

Divides a block into two halves.

$$L_{i+1} = R_i$$

$$R_{i+1} = L_i \text{ XOR } F_i(R_i)$$

The Engineering Genius

The internal function F_i **does NOT need to be invertible.**

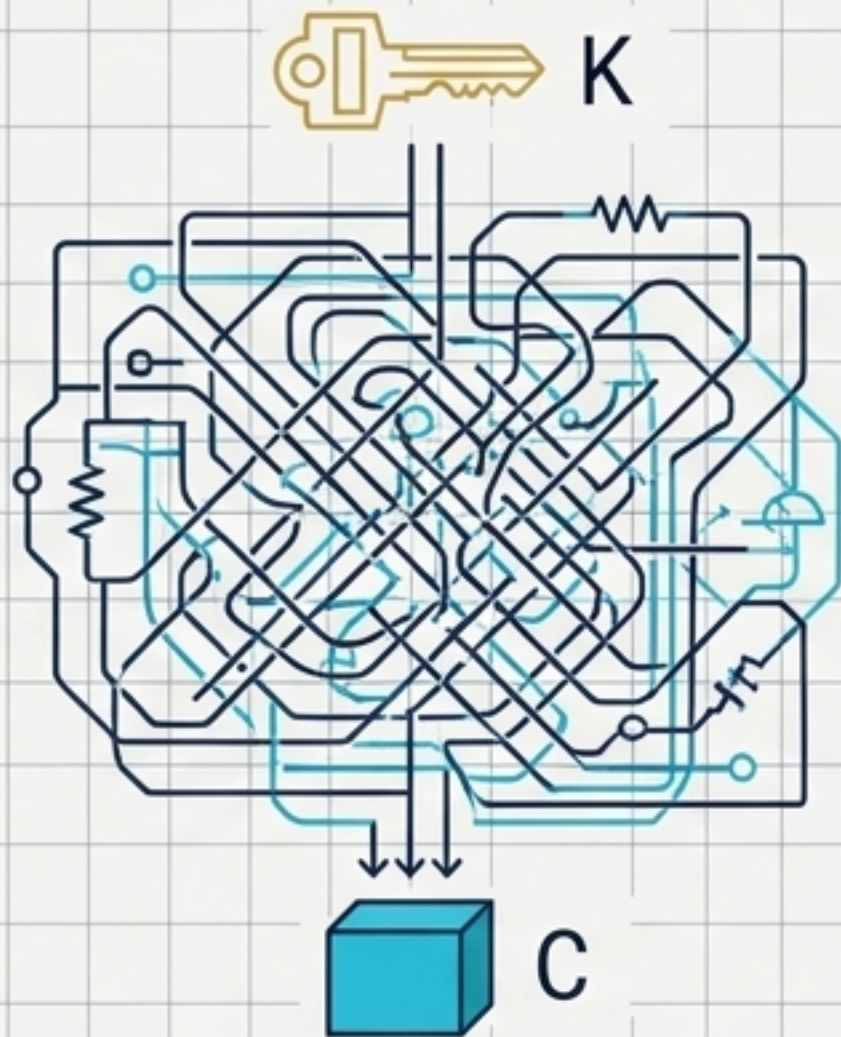
The overall transformation remains **reversible during decryption** because the structural XOR allows us to bypass it and compute:

$$L_i = R_{i+1} \text{ XOR } F_i(L_{i+1}).$$

This ingenious bypass is the foundation of DES.

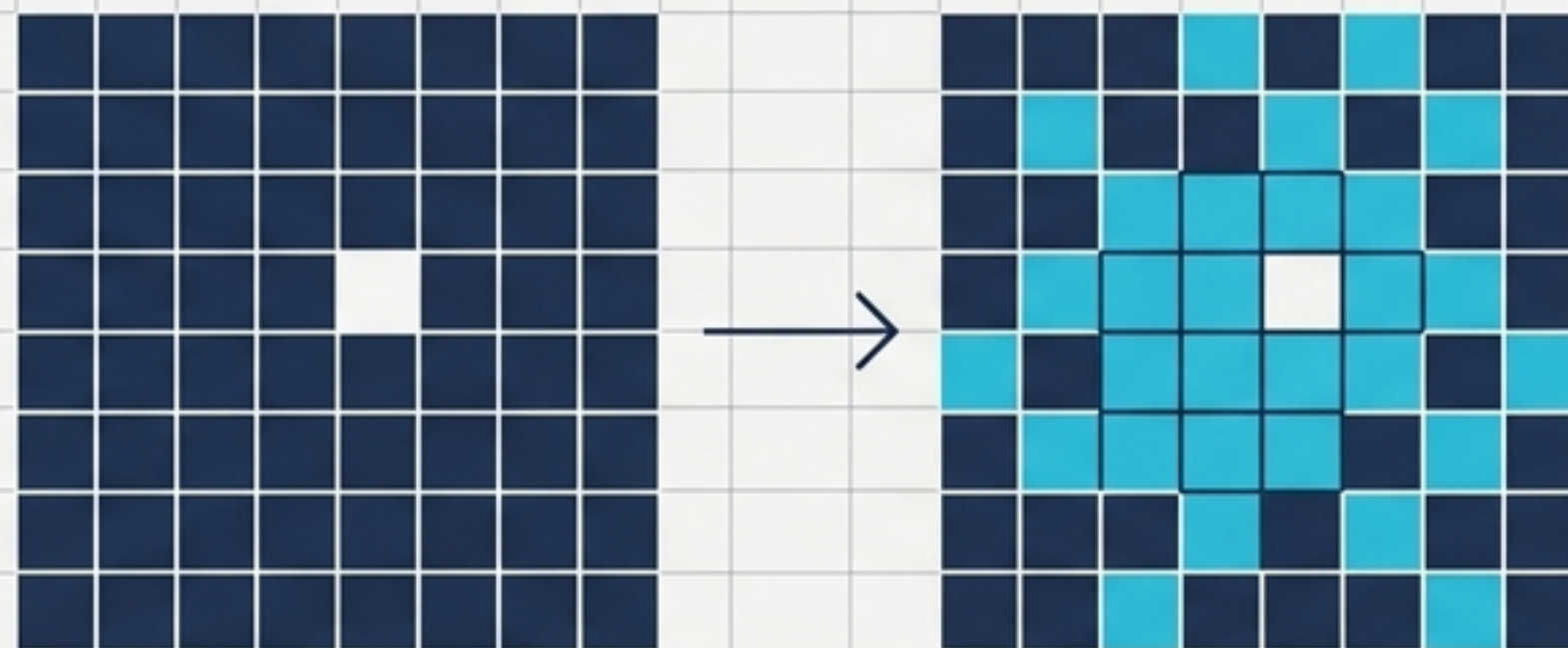
The Goals of Transformation: Confusion & Diffusion

Confusion



Complicates and obscures the mathematical relationship between the secret key and the ciphertext. Prevents attackers from deducing the key from ciphertext statistics.

Diffusion (The Avalanche Effect)



Spreads the influence of individual plaintext bits across the entire ciphertext. Flipping a single plaintext bit should change roughly 50% of the resulting ciphertext bits.

Block Cipher Generations

Cipher	Key Size	Architecture	Modern Status
DES	56-bit effective	64-bit block, 16-round Feistel	❌ Obsolete. Key space (2^{56}) fell to practical brute-force search as hardware improved.
3DES	112/168-bit	$C = E_{K_3}(D_{K_2}(E_{K_1}(M)))$	⚠️ Deprecated. Slow, vulnerable to meet-in-the-middle attacks, and limited by a 64-bit block size.
AES	128/192/256-bit	128-bit block, Sub-Perm network over $GF(2^8)$	✅ Global Standard. Selected by NIST (Rijndael). Relies on public construction and extensive cryptanalysis.

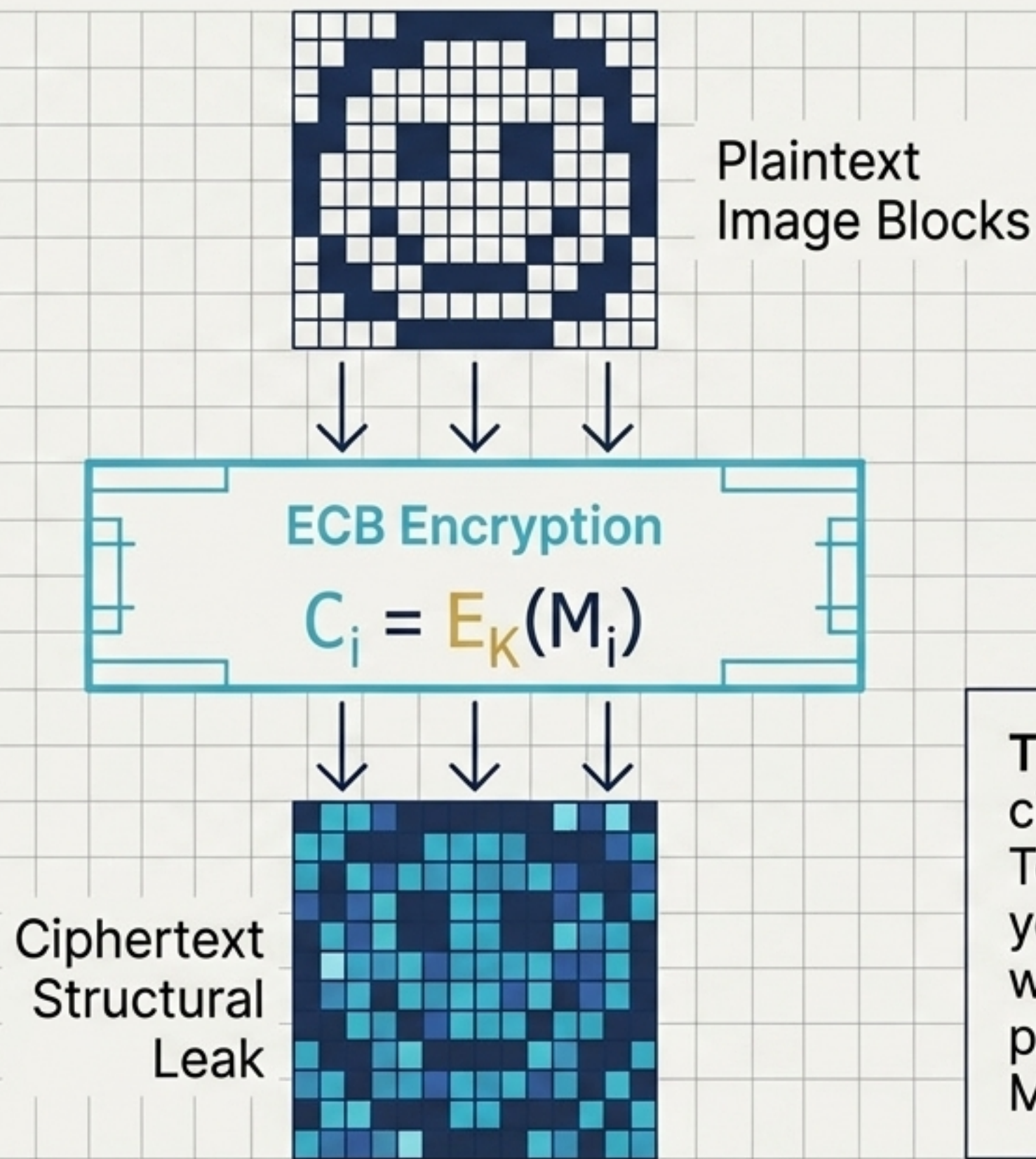
Why Algorithms Alone Fail: The ECB Leak

The Flaw of Electronic Codebook (ECB):

ECB encrypts each block independently. Therefore, identical plaintext blocks yield identical ciphertext blocks ($M_i = M_j \Rightarrow C_i = C_j$).

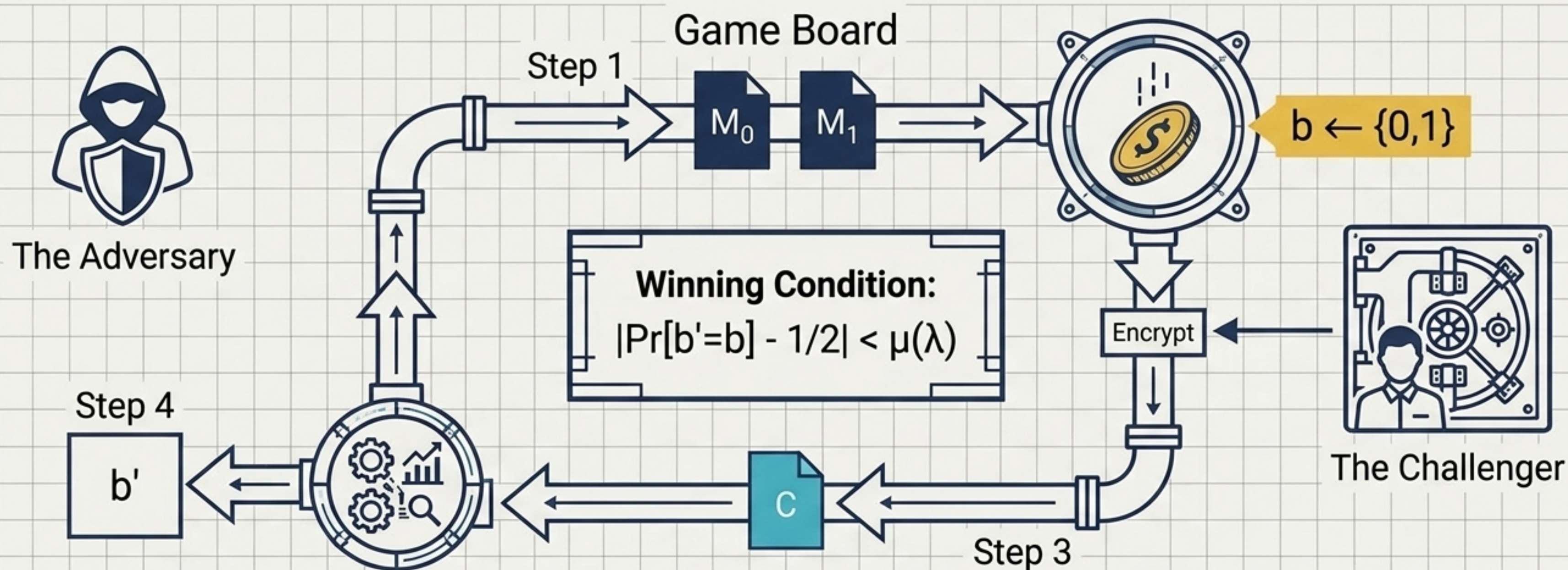
The Flaw of Electronic Codebook (ECB):

ECB encrypts each block independently. Therefore, identical plaintext blocks yield identical ciphertext ($M_i = M_j \Rightarrow C_i = C_j$).



The Takeaway: A block cipher is just a permutation. To encrypt messages safely, you must combine the cipher with nonces, counters, and previous ciphertexts using a Mode of Operation.

Playing the Security Game: IND-CPA



Indistinguishability (IND-CPA):

Security is proven through an explicit adversarial game modeling real-world influence over plaintext.

The Winning Condition:

An efficient adversary's advantage over a random guess must be negligible.

The Result:

Requires Randomized Encryption. Encrypting the same plaintext twice must produce different ciphertexts: $\text{Enc}_K(M; r_1) \neq \text{Enc}_K(M; r_2)$.

Modes of Operation: The Diagnostic Matrix

	Formula	Attributes	Verdict
ECB (Electronic Codebook)	$C_i = E_K(M_i)$	• Fails Confidentiality (leaks structure). • No Integrity.	 Status: Do Not Use.
CBC (Cipher Block Chaining)	$C_i = E_K(M_i \text{ XOR } C_{i-1})$	• Provides Confidentiality. • No Integrity. • Requires unpredictable IV. • Not parallelizable.	 Status: Legacy / Constrained.
CTR (Counter Mode)	$C_i = M_i \text{ XOR } E_K(\text{ctr}_i)$	• Turns block cipher into stream cipher. • Parallelizable. No Integrity. • Catastrophically fails if counter is reused.	 Status: Secure but Fragile.
GCM (Galois/Counter Mode)	Integrates CTR confidentiality with $GF(2^{128})$ arithmetic tag.	• Provides both Confidentiality and Integrity. • Highly parallelizable.	 Status: Modern Standard.

The Grand Synthesis: Authenticated Encryption

